CSS Lab
Computer System Security Lab
KENTECH - Energy AI Institute

HD-NIDSBench: 고차원 컴퓨팅을 활용한 효율적인 침입탐지 시스템에 대한 벤치마킹 프레임워크

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서론

1. The Challenge

Attack Surface: IoT expansion increases vulnerability.
Structural Risk: One weak device threatens the network.
Edge Constraints: Limited memory/ compute on gateways.

2. Deep Learning Limits

Why DL fails at the Edge:

- High computational cost
- Complex backpropagation
- Slow training on low-spec hardware

3. The Solution: HDC

Hyperdimensional Computing

⚡ **Lightweight**
Fast & Low Memory

🛡️ **Robust**
Noise Resistant

🧠 **Simple**
No Backprop

- HD-NIDSBench: Performance Analysis Framework
- Tested on Benchmark Datasets:

  Mirai, NSL-KDD, UNSW-NB15

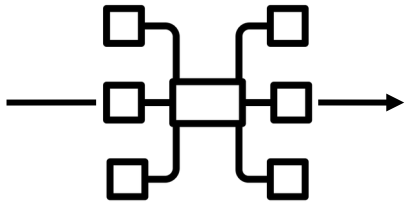
이론적 배경 : Hyperdimensional Computing



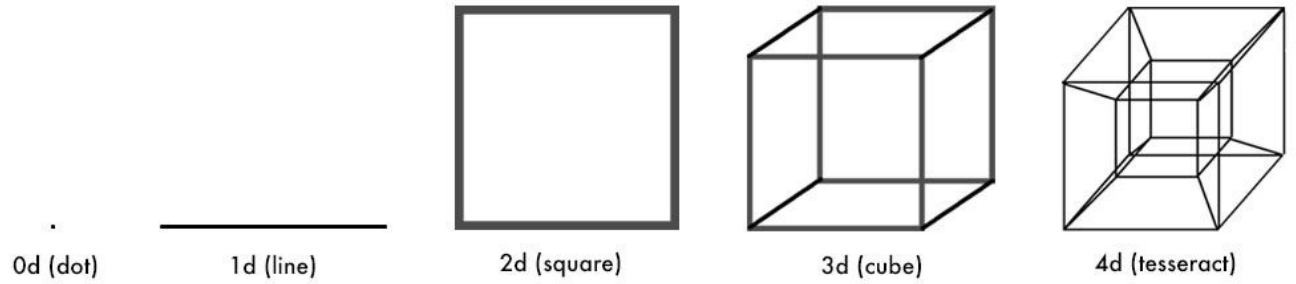
Brain-inspired

Hypervector: 10000 차원 이상의 거대한 벡터

[Feature A]
1dimension

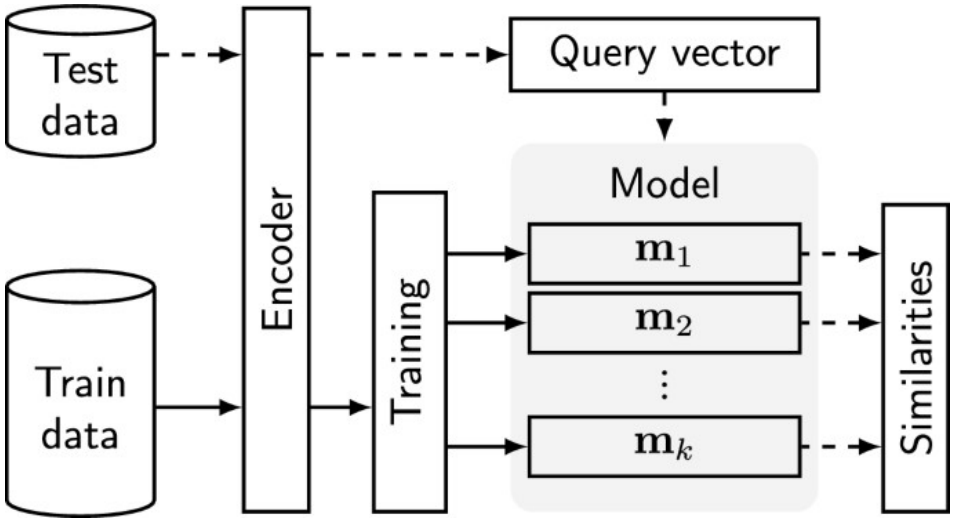


1 0 1 1 0 0 1 0 ... (10,000 bits) ... 1 0 1
10000 dimension



이론적 배경 : Hyperdimensional Computing

- HDC 의 장점
- Holographic: 모든 정보를 하이퍼벡터 (10,000 차원) 에 걸쳐 분산하여 저장
- Robustness: 일부 요소에 오류가 발생하더라도 전체 정보에 큰 손실이 없음
- Effectiveness: 단순하고 병렬 처리에 유리한 비트 연산을 사용해 속도와 메모리 효율성



Encoding 입력 데이터 ▯ 고차원 공간으로 매핑

Training

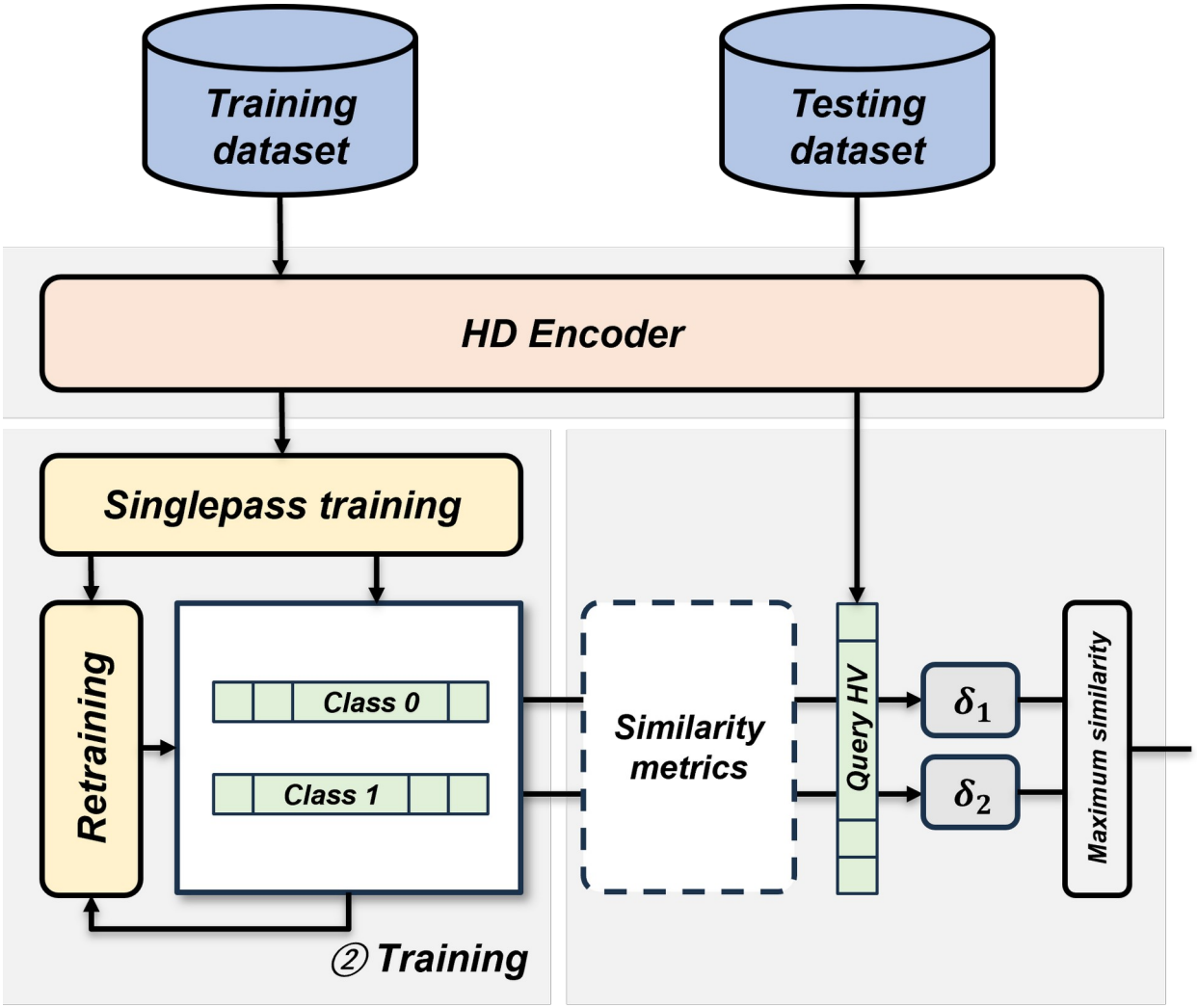
- Single pass: $C_l = \sum_{j=1}^{J} \mathcal{H}_j^l$
- Retraining $C_l \leftarrow C_l + \eta[1 - \delta(\mathcal{H}, C_l)] * \mathcal{H}$

$$C_{l'} \leftarrow C_{l'} + \eta[1 - \delta(\mathcal{H}, C_{l'})] * \mathcal{H}$$

Inference

1) Dot metric
2) Cos metric
3) Geodesic distance
4) Fisher-Rao metric
5) Tanimoto coefficient

HD-NIDSBench 설계



① Encoding
- ID-Level
- Random Projection
- Nonlinear

③ Inference
- Dot metric
- Cos metric
- Geodesic distance
- Fisher-rao metric
- Tanimoto efficient

HD-NIDSBench 평가

- 실험 세팅
  - 데이터셋

		normal	benign	total	features	number of devices
Mirai	training	38586	2307	40893	48	
	test	8587	748	9335		
NSL-KDD	training	67343	58630	125973	41	11
	test	9711	12833	22544		
UNSW-NB15	training	56000	119341	175341	34	45
	test	37000	45332	82332		

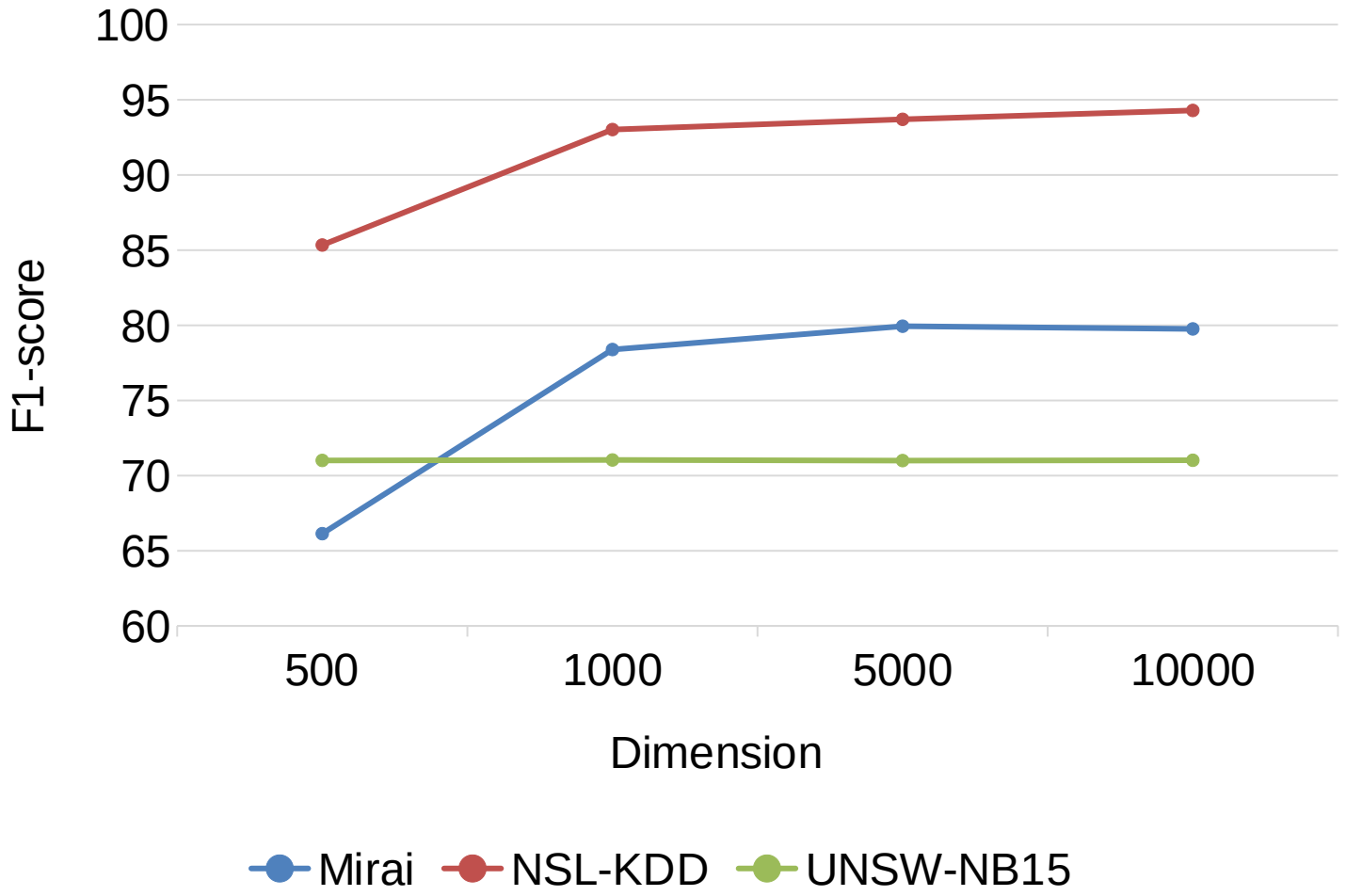
  - 수치형 및 범주형 특징 , Min-max scaling
  - Python 라이브러리 hdzoo 기반 벤치마크 , AI analyzer 활용 성능 비교

+ LightGBM, Random Forest, Feed-forward, Naïve Bayes

Evaluated by F1-score

HD-NIDSBench 평가

- 차원 크기에 따른 성능 평가



Control Variable	
Encoding	Nonlinear
Similarity	Cosine
Training	Retraining

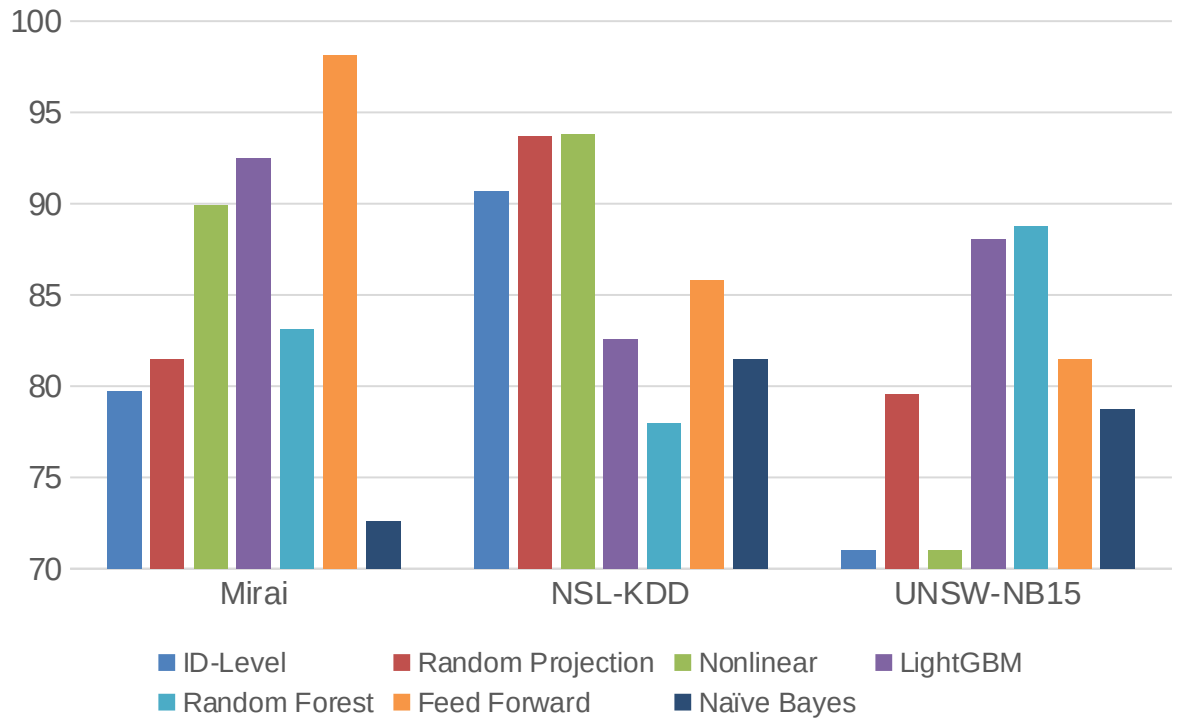
Independent Variable	
Dimension	500, 1000, 5000, 10000
Dataset	Mirai, NSL-KDD, UNSW-NB15

 차원 수를 높일수록 F1 score 가 점진적으로 향상

HD-NIDSBench 평가

- 데이터별 성능 평가

		Mirai	NSL-KDD	UNSW-NB15
	Retrain	F1score	F1score	F1score
ID-level	False	61.92	73.82	69.69
	True	79.76	90.74	71.02
Random Projection	False	61.92	79.89	73.19
	True	81.58	93.73	**79.57**
Nonlinear	False	69.74	76.09	79.42
	True	**89.97**	**93.82**	71.0
LightGBM		92.53	82.57	88.06
Random Forest		83.12	77.98	88.78
Feed Forward		98.11	85.82	87.15
Naïve Bayes		72.6	81.5	78.76



Control Variable	
Dimension	10000
Similarity	Cosine

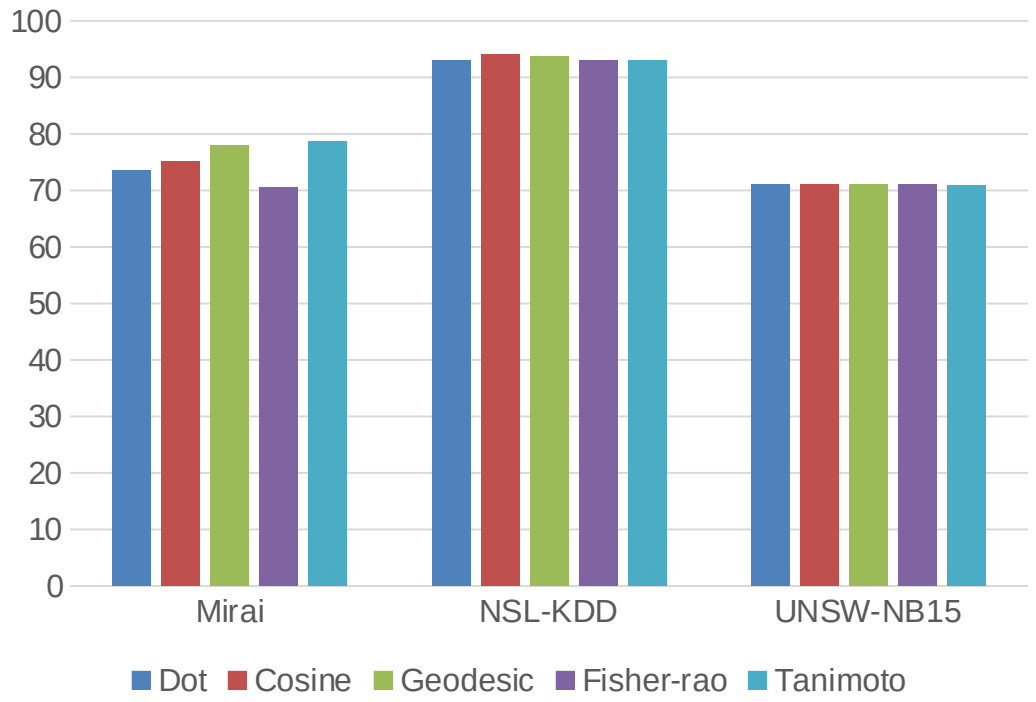
Independent Variable	
Dataset	Mirai, NSL-KDD, UNSW-NB15
Encoding	ID-level, Random projection, Nonlinear, LightGBM, Random Forest, Feed Forward, Naïve Bayes
Training	Retraining, Singlepass

- Mirai 와 NSL-KDD 에서는 Nonlinear 인코딩이 , UNSW-NB15 에서는 Random Projection 이 우수한 성능
- Retrain 을 적용한 모델이 평균 **16.95%p** F1-score 향상

HD-NIDSBench 평가

- 데이터별 성능 평가

		Mirai	NSL-KDD	UNSW-NB15
	Sim metrics	F1score	F1score	F1score
ID-level	Dot	77.31	92.23	71.02
	Cos	78.78	92.58	71.02
	Geodesic	96.95	91.63	71.02
	Fisher-rao	87.98	89.69	71.02
	Tanimoto	92.23	91.68	71.02
Random Projection	Dot	83.37	94.53	78.41
	Cos	86.66	93.11	79.31
	Geodesic	81.72	91.92	77.72
	Fisher-rao	84.95	96.04	77.42
	Tanimoto	82.14	92.5	77.94
Nonlinear	Dot	73.54	92.93	71.01
	Cos	75.13	94.08	71.02
	Geodesic	77.87	93.73	71.01
	Fisher-rao	70.6	93.0	71.02
	Tanimoto	78.58	93.02	70.96



Control Variable	
Dimension	10000
Training	Retraining

Independent Variable	
Dataset	Mirai, NSL-KDD, UNSW-NB15
Encoding	ID-level, Random projection, Nonlinear
Similarity	Dot, Cosine, Geodesic, Fisher-rao, Tanimoto

- 유사도 별 성능의 **표준편차 = 0.84, CV < 1.0%** 로 성능 차이가 거의 발생하지 않음
- 반복적인 재학습이 효과적이라는 의미

HD-NIDSBench 평가

- 메모리 효율성 평가

		Mirai	NSL-KDD	UNSW-NB15
	Retrain	F1score	F1score	F1score
ID-level	False	61.92	73.82	69.69
	True	79.76	90.74	71.02
Random Projection	False	61.92	79.89	73.19
	True	81.58	93.73	79.57
Nonlinear	False	69.74	76.09	79.42
	True	89.97	93.82	71.0
LightGBM		92.53	82.57	88.06
Random Forest		83.12	77.98	88.78
Feed Forward		98.11	85.82	87.15
Naïve Bayes		72.6	81.5	78.76

HDC	2.44KB
Feed Forward	193KB

- HDC 의 메모리 효율성이 **80 배** 가까이 우수함

결론 및 제언

HD-NIDSBench Results

- +17% Performance Boost via Retraining
- 80x Higher Memory Efficiency vs Neural Networks
- Proven Practicality for IoT Edge Devices

Future Directions

- Multi-class Classification: Identify specific attack categories.
- Anomaly Detection: Recognize unknown/zero-day attacks.
- Data Preprocessing: Resolve class imbalance issues.